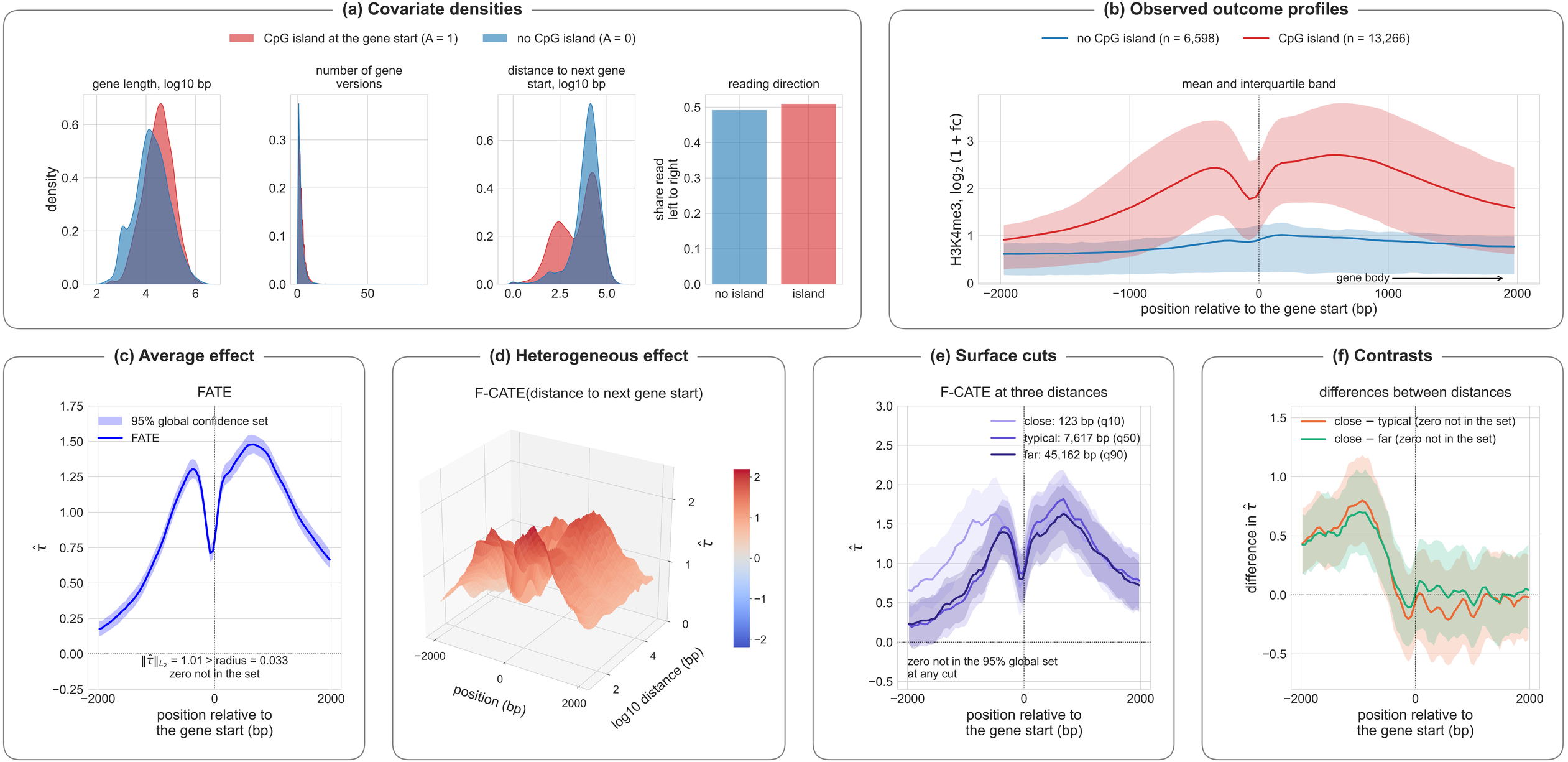

(a) Covariate densities
CpG island at the gene start (A = 1)
no CpG island (A = 0)
gene length, log10 bp
number of gene versions
distance to next gene start, log10 bp
reading direction
density
share read left to right
no island
island
(b) Observed outcome profiles
no CpG island (n = 6,598)
CpG island (n = 13,266)
mean and interquartile band
H3K4me3, log2 (1 + fc)
gene body
position relative to the gene start (bp)
(c) Average effect
FATE
95% global confidence set
FATE
‖τ̂‖L2 = 1.01 > radius = 0.033
zero not in the set
position relative to the gene start (bp)
(d) Heterogeneous effect
F-CATE(distance to next gene start)
position (bp)
log10 distance (bp)
(e) Surface cuts
F-CATE at three distances
close: 123 bp (q10)
typical: 7,617 bp (q50)
far: 45,162 bp (q90)
zero not in the 95% global set at any cut
position relative to the gene start (bp)
(f) Contrasts
differences between distances
close − typical (zero not in the set)
close − far (zero not in the set)
difference in τ̂
position relative to the gene start (bp)